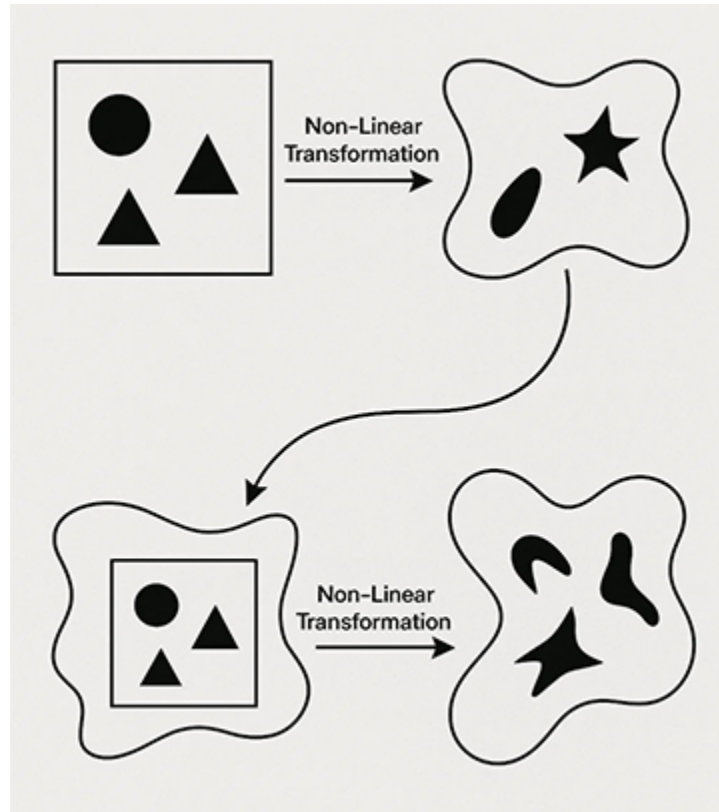


Addendum A:

Non-Linear Transformations for the VCLI-G Space

(Anisotropic Power Equalization + Residual Contrast Expansion)



Abstract

This Addendum formalizes the non-linear transformation framework that extends the Visual Cognitive Load Index (VCLI-G) into a perceptually normalized space. It introduces two complementary operators: **Anisotropic Power Equalization (APE)** and **Residual Contrast Expansion (RCE)**, these together construct a distribution-aware perceptual manifold above the canonical geometric measure (CANON).

APE applies axis-specific power-law scaling to equalize perceptual step size across the Structural Coherence Index (SCI) $\times$ VCLI-G plane, analogous to divisive gain control or γ -linearization in visual neuroscience and color science. RCE then amplifies local residual variance within SCI bands, revealing within-tier strategic differences between resolution and sustained tension. The combination stabilizes higher-order moments (skewness, kurtosis) while preserving ordinal integrity, converting statistical distance into perceptual distance.

Within this three-stage architecture, **CANON** $\rightarrow$ **APE** $\rightarrow$ **RCE**, the system distinguishes absolute sophistication, perceptual robustness, and artistic strategy. The canonical layer ensures deterministic reproducibility; APE and RCE introduce controlled contextual adaptivity that mirrors human perceptual recalibration. Together they transform the VCLI-G $\times$ SCI field from a static metric into a responsive diagnostic model of visual intelligence.

$\rightarrow$ For axis definitions and quadrant interpretation, see 'What the VCLI-G and SCI Measure.

What the VCLI-G and SCI Measure

The Visual Cognitive Load Index (VCLI-G) and the Structural Coherence Index (SCI) form the two primary coordinates of the Visual Thinking Lens diagnostic field. Each image, frame, or generative output is represented as a point in this space:

- **VCLI-G (x-axis):** measures *perceptual strain*, how much visual or cognitive work the composition demands from the viewer before it resolves into sense. High VCLI-G corresponds to images that delay closure, contain dense mark systems, or provoke sustained interpretive attention. Low VCLI-G indicates quick readability, minimal strain, or deliberate emptiness.
- **SCI (y-axis):** measures *structural coherence*, how effectively the visual system maintains internal order, rhythm, and balance under load. High SCI reflects strong compositional scaffolding or formal discipline; low SCI marks collapse, fragmentation, or unresolved spatial logic.

Together these axes define a **two-dimensional perceptual field** where compositional behavior can be located and compared.

Plots and quadrants are oriented as follows:

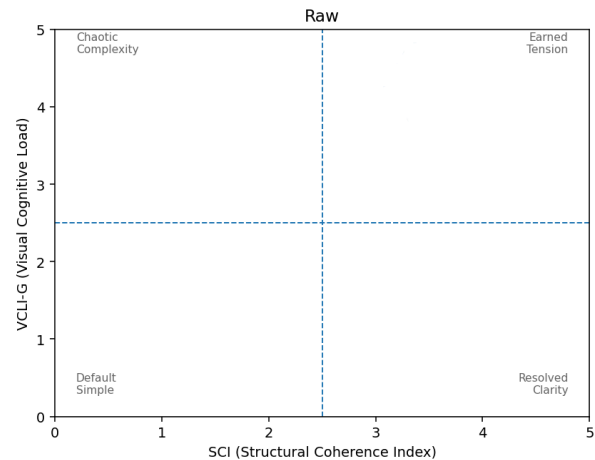


Figure Caption: VCLI-G × SCI Quadrant Map

Each image is plotted by its Visual Cognitive Load Index (VCLI-G, horizontal) and Structural Coherence Index (SCI, vertical). The resulting field visualizes how perceptual tension interacts with compositional order. Movement to the right increases cognitive load (more visual work before resolution); upward movement increases structural coherence (stronger internal logic). The upper-right quadrant, **Earned Clarity**, marks the highest form of visual intelligence: compositions that sustain tension yet remain structurally resolved. Diagonal drift across this field reveals an image’s developmental arc, from effortless simplicity through chaotic saturation toward deliberate equilibrium.

Horizontal axis: VCLI-G (Cognitive Load)→ VCLI-G (Cognitive Load)→ Vertical axis: SCI (Structural Coherence)↑

Quadrant	Characterization	Typical Reading
Upper-Right (High SCI, High VCLI-G)	<i>Earned Clarity</i>	Stability achieved through tension and complexity, mastery under strain.
Lower-Right (Low SCI, High VCLI-G)	<i>Chaotic Complexity</i>	Overloaded systems where tension overwhelms structure.
Upper-Left (High SCI, Low VCLI-G)	<i>Resolved Clarity</i>	Ordered, tranquil images that resolve efficiently.
Lower-Left (Low SCI, Low VCLI-G)	<i>Default Simplicity</i>	Undeveloped or neutral states with little structural or perceptual demand.

Interpreting motion within this field allows evaluators to trace *how coherence and cognitive load evolve together*, whether an image gains stability through refinement, risks collapse under excess, or finds deliberate equilibrium through restraint.

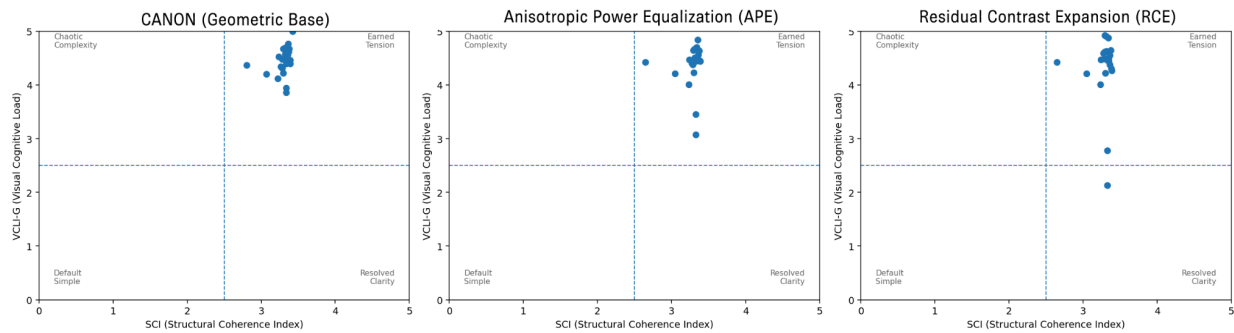
Figure Caption: VCLI-G × SCI Quadrant Map with Transformations

The canonical VCLI-G × SCI field (center) shows the unwarped relationship between perceptual load and structural coherence.

When **Anisotropic Power Equalization (APE)** is applied, the horizontal (VCLI-G) axis undergoes differential gain expansion, stretching high-load values and compressing low ones. This widens separation among dense, tension-rich works without disturbing ordinal rank. The field appears to “breathe” horizontally, clarifying differences between chaotic excess and earned structural control.

The subsequent **Residual Contrast Expansion (RCE)** acts locally within SCI bands, amplifying vertical dispersion where subtle compositional variances are hidden. This creates the characteristic “waterfall” or “fan” effect: each SCI corridor opens slightly, revealing fine gradations of perceptual strain. RCE does not change quadrant identity, it accentuates micro-relationships, exposing how stability and load fluctuate within coherent regimes.

Together, APE and RCE re-scale the field from an objective grid into a **perceptually balanced manifold**, where visible distance approximates how the eye and mind experience structural and cognitive tension.



Each point represents one image. Rightward motion = higher perceptual load (VCLI-G); upward motion = stronger structural coherence (SCI). Read the field as a behavioral map: images travel diagonally toward **Earned Clarity** as they learn to hold tension without collapse. APE widens this map for perceptual accuracy; RCE reveals fine texture within each coherence band.

To be clear, APE and RCE aren’t purely *per-image* transforms. They rely on **distribution-aware normalization**: global statistics computed from the *entire batch* that is fed to the notebook. That means each image’s *transformed* value (and occasionally its normalized pre-value) depends on the **context of the full dataset** with its mean, variance, and percentile structure.

1. Canonical layer vs. transformed layer

Stage	What happens	Scope
Canonical VCLI-G computation	feature-wise scoring (entropy, void ratio, etc.) combined via fixed weights	Independent per image
Normalization / scaling	typically z-score or percentile normalization across the batch	Global
APE / RCE transforms	use percentiles, local fits, or bin means derived from the whole batch	Global (distribution-dependent)

Thus, if changed: the **number of images**, the **order** (which shifts how `pd.qcut` or percentile bins slice the data), or include even **one high-load or low-load outlier**, it will alter the reference distribution, and thus the mapped values (especially in APE and RCE). Think of the APE/RCE layer as a *contextual mirror*: it only knows what “high load” or “low coherence” means **relative to its peers in the current run**. If the population changes, the perception of “extreme” changes.

Interpretive Stability: Why the Field Moves and Why it is Important

The VCLI-G and SCI system is stable in computation but adaptive in interpretation.

Each image's canonical values remain fixed, yet their **relative placement** in the perceptual field can shift as new works enter the batch.

This apparent volatility arises because both APE (Anisotropic Power Equalization) and RCE (Residual Contrast Expansion) operate on **ensemble statistics**; they normalize perceptual scale to the distribution currently under observation. Adding or removing works changes the field's mean and variance, prompting the system to re-center its sense of "average load" and "typical coherence."

Rather than a flaw, this behavior mirrors the way human perception recalibrates contrast and salience in different viewing contexts: a Corot and a Cézanne seen alone seem comparable; surrounded by mid-tone academic painting, their difference becomes categorical. The model's elasticity therefore reflects **contextual fidelity**, its ability to model perception as a relational act rather than an absolute measure. Viewers recalibrate what "complex" means based on what they've just seen. That's why VCLI-G's transforms are deliberately context-sensitive: they model *relative cognitive load* rather than an absolute one.

In practice, canonical scores provide deterministic reproducibility, while the dynamic APE/RCE views express *interpretive stability*: the capacity of the metric to remain coherent even as its frame of reference evolves. Under this warped view, clusters once tightly packed in the center separate into intelligible trajectories, making the diagram a living map of compositional evolution rather than a static scatter plot.

While this context sensitivity can frustrate reproducibility, it's actually an **intentional feature** for exploratory visualization, it lets the field "breathe" differently when you mix AI, human, or hybrid images. The canonical numbers (raw VCLI-G, SCI) remain absolute; it's only the *perceptual normalization layer* that re-interprets them in context.

Design takeaway

This is: **Volatility = sensitivity to context. That's what makes the VCLI-G/SCI plane a living perceptual model rather than a static metric.**

The VCLI-G index adapts to ensemble statistics: as the field broadens, distances re-encode perceptual rarity. This volatility is perceptual fidelity, not instability.

1. Mathematical Basis: Anisotropic Power Equalization (APE)

APE defines a differential-gain transform operating anisotropically on SCI and VCLI-G components as follows:

To better visualize and compare samples in the Structural Coherence Index (SCI) × Visual Cognitive Load Index (VCLI-G) plane, we introduce a controlled non-linear remapping termed **Anisotropic Power Equalization (APE)**.

APE performs differential γ -scaling along each perceptual axis to normalize perceptual sensitivity:

$$z_i = \frac{x_i - \mu_i}{\sigma_i}, \quad z'_i = \text{sign}(z_i)|z_i|^{\gamma_i}, \quad x_{w,i} = \mu_i + \sigma_i z'_i,$$

with exponents

$$(\text{SCI}', \text{VCLI-G}') = (\text{sign}(\text{SCI})|\text{SCI}|^{\gamma_s}, \text{sign}(\text{VCLI-G})|\text{VCLI-G}|^{\gamma_v}) \quad \text{with } \gamma_v > \gamma_s$$

This operation implements a **non-linear gain control** analogous to the *Naka–Rushton* or *contrast-response* function used in retinal and cortical models

$$R = R_{\max} I^n / (I^n + \sigma^n)$$

(Heeger 1992; Carandini & Heeger 2012).

Low-magnitude deviations (minor tensions) are compressed, while high-magnitude deviations (intense perceptual strain) are expanded.

Mathematically, APE defines an anisotropic metric:

$$d_w^2 = (\Delta \text{SCI})^{2\gamma_{\text{SCI}}} + (\Delta \text{VCLI})^{2\gamma_{\text{VCLI}}},$$

producing a **differential-gain normalization** that spreads the dense central region of the perceptual field without disturbing rank order.

APE therefore approximates a **perceptual linearization**, similar to how CIELAB or IPT introduce nonlinear exponents to equalize perceptual step size.”

APE corresponds to *Option B (physical_neutral)* in code.

2. Residual-Based Local Variance: Residual Contrast Expansion (RCE)

Following APE, a local residual-amplification step (RCE) may be applied to each SCI band: **Residual Contrast Expansion (RCE)** to reveal hidden stratification inside coherence bands.

For each local SCI bin j :

$$y'_{ij} = \hat{y}_{ij} + k(y_{ij} - \hat{y}_{ij}), \quad k > 1,$$

where y_{ij} are VCLI-G values within SCI bin j

$\hat{y}_{ij}$ is the fitted local mean or trend, and k is the residual-gain constant ($k \approx 1.6$!approx!1.6k≈1.6).

This scales the residuals (variance amplification) while preserving each band's mean and slope, equivalent to **local heteroscedastic scaling** or **adaptive gain modulation** used in HDR tone-mapping (Reinhard et al. 2002).

Typical settings: five bins, $k \approx 1.6$ $k \approx 1.6$

RCE corresponds to *residual-gain mode* ($k \approx 1.6 - 1.8$ $k \approx 1.6 - 1.8$ $k \approx 1.6 - 1.8$)

3. Rationale and Empirical Grounding

Raw (SCI, VCLI-G) distributions tend to cluster near perceptual equilibrium ($\approx \text{SCI } 3 \pm 0.3$, $\text{VCLI-G } 3 \pm 0.4$), masking structural distinctions.

APE + RCE address this by:

- equalizing perceptual density across scales (APE), and
- amplifying within-band micro-variance (RCE).

Comparable mechanisms appear in:

- **Visual neuroscience:** divisive normalization and local gain control (Heeger 1992; Carandini & Heeger 2012).
- **Color science:** CIE Lab and IPT γ -scaling for perceptual uniformity.
- **Imaging:** tone-mapping and adaptive histogram equalization for local contrast.
- **Machine perception:** feature-space whitening and power-law embeddings for stability under heteroscedastic noise.

Together, these justify non-linear equalization as perceptually faithful and statistically stabilizing.

RCE produces a **derived visualization space**; canonical VCLI-G values must remain unwarped for quantitative reporting.

4. When and Where to Use

Recommended Use (AP + RCE View):

- Cross-modality comparison (e.g., AI-generated vs human imagery).
- Exploratory visualization of within-band cognitive-load dispersion.
- Presentation contexts requiring perceptual interpretability rather than raw metric fidelity.

Raw View (Linear Space):

- Quantitative regression, calibration, or algorithmic scoring.
- Validation against psychophysical ground truth.

RCE should remain **diagnostic only**, it modifies variance representation, not canonical index values.

5. Interpretive Summary

APE + RCE constitute a two-stage perceptual-metric transformation:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad (\text{SCI}, \text{VCLI}) \mapsto (\text{SCI}_w, \text{VCLI}_w),$$

where APE performs **global differential-gain equalization** and RCE performs **local residual-contrast amplification**.

$$x' = f_{RCE}(f_{APE}(x)) = f_{APE}(x) + k(f_{APE}(x) - \hat{f}_{APE}(x))$$

APE + RCE stabilize higher-order moments (skewness, kurtosis) across cohorts, improving regression interpretability. APE and RCE are post-hoc visualizations; the canonical dataset remains unaltered for regression and validation.

Formal notation for composition:

$$f_{RCE \circ APE}(x) = f_{RCE}(f_{APE}(x)) = \hat{x} + k(f_{APE}(x) - \hat{x})$$

The resulting field can be visualized as a **waterfall projection**, an anisotropic manifold where global normalization (APE) sets slope and local variance (RCE) determines rippling amplitude.

The resulting “waterfall” projection expresses latent manifold structure, with distinct attractor regions for *Default Simple*, *Chaotic Complexity*, *Earned Tension*, and *Resolved Clarity*; with perceptual spacing proportional to cognitive salience.

Terminology and Code Mapping

Framework Term	Purpose	Analog / Citation	Prior Code Term
Anisotropic Power Equalization (APE)	Axis-specific γ -scaling (global gain normalization)	Naka–Rushton law (1966); Heeger (1992)	Option B / <code>preset="physical_neutral"</code>
Residual Contrast Expansion (RCE)	Local residual variance amplification	Reinhard et al. (2002) tone-mapping; Carandini & Heeger (2012) gain control	<code>finger="residual"</code> (+ strength)
Differential Gain Warp (<i>synonym</i>)	Conceptual description for APE + RCE combination	Standard non-linear response normalization	—

Key References

- Naka, K. & Rushton, W. (1966). *S-potentials from luminosity units in retina*. *J Physiol* 185, 587–599.
- Heeger, D. J. (1992). *Normalization of cell responses in cat striate cortex*. *Visual Neuroscience* 9, 181–197.
- Carandini, M. & Heeger, D. (2012). *Normalization as a canonical neural computation*. *Nature Reviews Neuroscience* 13, 51–62.
- Reinhard, E. et al. (2002). *Photographic tone reproduction for digital images*. *ACM TOG* 21(3), 267–276.

Quick Decision Tree

Which view (examples)?

- Comparing across datasets/models? → Use **CANON** (absolute baseline)
- Visualizing AI vs. human work? → Use **APE** (spreads the AI cluster for visibility)
- Studying artistic strategy within masters? → Use **RCE** (reveals resolved vs. sustained)
- Studying iteration in AI imagery or physical imager? → Use **ALL** (reveals baseline, resolved vs. sustained, interpretive)
- Publishing quantitative results? → Report **CANON** (reproducible), show **APE/RCE** (interpretive)

Summary:

APE and RCE provide mathematically principled, perceptually grounded transformations that convert the raw VCLI-G field

into an interpretable, gain-normalized space, preserving ordinal integrity while revealing latent structural and cognitive dynamics within complex image sets.

MINI Translation Key Case Example



→ AI Portrait, 2025, Parrish; Reverie, Antoine, 1860–65, Camille Corot Dominique Sauveur Aubert (born 1817), the Artist's Uncle, as a Monk, 1866, Paul Cézanne; View near Rouen, 1825, Richard Parkes Bonington. All Museum work: The Metropolitan Museum of Art

The Three-Stage Measurement Architecture

CANON (Geometric Base): Absolute Geometric Measurement

What it measures: Total perceptual work demanded by the image

- **The question:** "How much geometry is here? How sophisticated is this in absolute terms?"
- **The mechanism:** Raw summation of geometric signals (G1-G4: centroid wander, void topology, curvature torque, occlusion entropy)
- **What it reveals:** Whether sophistication exists at all - the baseline complexity threshold
- **The scale:** 0-5+ (in practice: AI = 2.3-3.2, Working Artists = 2.8-3.8, Masters = 3.8-5.0+)

Output: Objective measurement - "This image contains X units of geometric demand"

Examples:

- Corot: 3.858 → "Sophisticated, well above AI baseline"
- Cézanne: 4.621 → "Highly sophisticated, master-level complexity"
- Bonington: 5.0 → "Peak sophistication, maximum geometric demand"
- AI: 2.6 → "Competent echo, default complexity"

Purpose: Establishes the tier. Separates masters from defaults, earned from emergent.

→ Results below are transforms and are distribution-dependent; absolute comparisons must use canonical values.

APE: Perceptually Uniform Spacing

What it measures: Perceptual distance between images (not just geometric distance)

- **The question:** "Is this sophistication robust, or just an artifact of data clustering? Does small geometric movement represent small or large perceptual change?"
- **The mechanism:** Differential power-law transform ($\gamma = 1.2$ for SCI, 2.0 for VCLI-G) that spreads the dense center and preserves the edges
- **What it reveals:** Whether differences are signal or noise - which distances are perceptually meaningful
- **The scale:** Same 0-5 range, but with perceptually uniform spacing (like CIE Lab does for color)

Output: Subjective-to-objective transform - "Here's where things sit in *perceived* complexity space"

Examples:

- Corot: 3.858 → 3.875 (0.4% shift) → "Already at field edge, robust"
- Cézanne: 4.621 → 4.742 (2.6% shift) → "Far from center, stable position"
- Bonington: 5.0 → 5.070 (1.4% shift) → "Extreme outlier, unmoved by spreading"
- AI cluster: 2.6 → 2.9 → 3.1 (spreads 20-30%) → "Dense center disperses for visibility"

Purpose: Makes cross-modal comparison valid. Ensures you're comparing like-to-like perceptually, not just numerically. Reveals whether canonical scores represent real perceptual separation or statistical artifact.

Why it matters: A 0.2 difference in the AI basket (2.6 vs 2.8) is perceptually trivial - they're all echoes. A 0.2 difference at the master tier (4.4 vs 4.6) is perceptually significant - different achievements. APE makes this explicit by spreading the first and preserving the second - **implementing the psychophysical principle that perceptual distance is not linear in stimulus magnitude (Stevens' Power Law, 1957).**

RCE: Contextual Stratification

- **What it measures:** Relative complexity within coherence bands
The question: "At this level of organization (SCI), is this image demanding or restrained compared to peers? Is sophistication achieved through sustained tension or through resolution?"
- **The mechanism:** Local residual variance amplification ($k \approx 1.6$) within SCI bins - scales deviations from the coherence-band trend line
- **What it reveals:** Artistic strategy - *how* the work achieves its demands, not just *that* it demands
- **The scale:** Rescaled 0-5 where position reflects strategy within peer group

Output: Contextual interpretation - "Given this level of coherence, here's the approach"

Examples:

- Corot: 3.858 → 2.128 (45% drop) → "Restrained relative to high-coherence peers - resolved mastery"
- Cézanne: 4.621 → 4.564 (1.2% hold) → "Sustained relative to high-coherence peers - earned tension"
- Bonington: 5.0 → 5.627 (12% rise) → "Exceptional even among high-coherence peers - comprehensive demand"
- AI: 2.6 → 2.8 (might rise 8%) → "High-variance within low-coherence band - but started too low to matter"

Purpose: Distinguishes artistic strategies within the masterwork tier. Separates "resolved through mastery" from "resolved by default," "earned tension" from "chaotic noise."

Why it matters: Canon (Geometric Base) tells you Corot (3.858) and AI (2.8) are different. RCE tells you Corot (→2.128) and Cézanne (→4.564) are *also* different - one resolves, one sustains. That's a distinction in artistic approach that raw geometric measurement can't capture.

How They Work Together: The Diagnostic Sequence

Stage 1 - CANON (Geometric Base): Does sophistication exist?

- **AI at 2.6:** No, below threshold → Default complexity
- **Corot at 3.858:** Yes, above threshold → Earned complexity
- **Cézanne at 4.621:** Yes, well above → Master-level complexity

Stage 2 - APE: Is that sophistication robust?

- **AI 2.6 → 3.0:** Spreads 15% → Was in dense cluster, now visible
- **Corot 3.858 → 3.875:** Shifts 0.4% → Already at edge, position confirmed
- **Cézanne 4.621 → 4.742:** Shifts 2.6% → Far from center, robust outlier

Stage 3 - RCE: What's the strategy at this coherence level?

- **AI 2.6 → 2.8:** Rises modestly → High variance in low-coherence band (but still default-tier)
- **Corot 3.858 → 2.128:** Drops 45% → Restrained strategy, atmospheric resolution
- **Cézanne 4.621 → 4.564:** Holds steady → Sustained tension, constructive pressure

Work	CANON	APE	RCE	Interpretation
AI Portrait	2.6	3.0 (+15%)	2.8	Default complexity
Corot	3.858	3.875 (+0.4%)	2.128 (-45%)	Resolved mastery
Cézanne	4.621	4.742 (+2.6%)	4.564 (-1.2%)	Sustained tension
Bonington	5.0	5.070 (+1.4%)	5.627 (+12%)	Comprehensive demand

Why all Three:

Canon (Geometric Base) alone would tell you:

- "Corot (3.858) and Cézanne (4.621) are both sophisticated"
- But can't tell you Corot resolves while Cézanne sustains

APE alone would tell you:

- "These scores are robust, not clustering artifacts"
- But can't tell you what the scores *mean* strategically

RCE alone would tell you:

- "Corot is restrained, Cézanne is sustained"
- But without Canon (Geometric Base), you wouldn't know both are masters (versus AI defaults)

Together, they deliver:

"Corot achieves master-level sophistication (Canon: 3.858) through atmospheric resolution (RCE: 2.128) - restrained relative to peers at his coherence level. Cézanne achieves even higher sophistication (Canon: 4.621) through sustained constructive tension (RCE: 4.564) - pressured even relative to peers. Both are robustly above the AI default cluster (APE confirms), but employ fundamentally different spatial strategies. An AI at 2.6 lacks the baseline sophistication (Canon) to reach either approach - it can't resolve through mastery because it never achieved mastery to begin with."

The System Distinguishes:

1. **Sophisticated vs. Default** (Canon (Geometric Base))
 - Masters (3.8-5.0) vs. AI (2.3-3.2)
2. **Robust vs. Artifact** (APE)
 - Real separation vs. clustering noise
3. **Resolved vs. Sustained** (RCE within sophistication tier)
 - Corot (restrained mastery) vs. Cézanne (pressured mastery)
4. **Mastery vs. Default** (RCE across tiers)
 - Corot drops to 2.128 but *from* 3.858 = "resolved through mastery"
 - AI might score 2.8 in RCE but *from* 2.6 = "resolved by default"

That fourth distinction - "resolved through mastery" vs. "resolved by default" - is what makes this a control architecture instead of just a scoring system.

It's the difference between:

- A minimalist Agnes Martin painting (low VCLI-G, high SCI, master-tier Canon (Geometric Base) → strategic resolution)
- A centered gradient with text overlay (low VCLI-G, high SCI, default-tier Canon (Geometric Base) → lazy default)

Both land in "Resolved Clarity" on the plane. But **the trajectory through Canon (Geometric Base) → APE → RCE tells you which is which.**

Summary

APE and RCE extend the canonical VCLI-G × SCI plane into a distribution-aware perceptual manifold. APE performs global γ -scaling to equalize perceptual step size; RCE amplifies residual variance within structural bands. Together they translate statistical distance into perceptual distance, revealing artistic strategies of resolution versus tension while preserving canonical reproducibility through stored reference stats. As this example illustrates, it translates from the geometric to artistic strategy → or as little as → this image is what it is.

Authorship

This framework was developed by Russell Parrish through iterative dialogue with GPT-5, Claude 3.7 Sonnet, and Gemini 2.0 Flash. The conceptual architecture, design decisions, and all substantive judgments are human-authored. The language models served as technical sounding boards, providing feedback on mathematical formulation, code implementation, statistical interpretation, and rhetorical clarity. Every structural choice, from the three-primitive decomposition to the TEL layer design, originated from artistic practice and was refined through multi-model consultation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, computational aesthetics, and perceptual theory.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

Appendix 1:
Context-relative rather than absolute.

Baseline Expectation: Determinism

Engineers expect that a metric is *stable under identical input* → *this is the canon (geometric base)*.

- Feed the same image twice → expect identical scores.
- Feed the same image set in a different order → expect no change (APE/RCE violate this, because normalization is global). So their first reaction is: “*Why does order or cohort matter?*”

This isn't a bug in design, it's a design philosophy clash: classical engineering values **deterministic comparability**; perceptual modeling values **contextual adaptivity**.

Research Expectation: Distribution Awareness

Researchers, especially those in perceptual modeling, psychophysics, or ML evaluation, *expect* a contextual layer to exist. “Load” and “coherence” are not absolute, but **relative to ensemble statistics**. Treat APE/RCE as *normalization stages*, not part of the metric core.

- **Separate raw measures** (canonical VCLI-G, SCI) from **distributional transforms** (APE/RCE visualizations).
- Export both: → raw = fixed, comparable; → warped = contextual, diagnostic.

Engineer Expectation: Reproducibility Switch

For the software engineer **toggle or mode flag** that freezes reference stats. This doesn't lose contextual richness; it means reproducibility can be forced on demand. For example:

```
ape_mode = "dynamic" # or "frozen"
rce_mode = "dynamic"
```

- *Dynamic* → context-sensitive, recomputes per batch (good for perception studies).
- *Frozen* → uses stored μ , σ , bins (good for regression, benchmarks).

That's the compromise between art-research and engineering.

Framed:

Layer	Type	Expectation	Analogy
Canonical (VCLI-G, SCI)	Deterministic metric	Fixed under repeat runs	RMS contrast, entropy, or SSIM in vision science
APE	Contextual warp	Recomputes to reflect <i>relative perceptual distance</i>	Adaptive gamma or gain control
RCE	Local diagnostic	Recomputes within bands to show <i>relative demand</i>	Local contrast normalization

So the expectation is **two-channel operation**:

- Channel A → reproducible absolute metrics
- Channel B → perceptual, context-aware diagnostic view

Practical Precedent

Precedent for this duality in every perceptual modeling field:

- **Color spaces**: CIE XYZ (absolute) vs. CIELAB (perceptually uniform, context-scaled).
- **Compression metrics**: PSNR (absolute) vs. SSIM (contextual).

- **Neural network normalization:** BatchNorm (distribution-aware) vs. LayerNorm (frozen reference).
- **Audio loudness:** SPL (absolute) vs. LUFS (perceptual, time-weighted).

Researchers document contextual for paper, docstring, or GUI clearly state: *“Transforms are distribution-dependent; absolute comparisons must use canonical values.”*

Persona	Reaction to APE/RCE behavior
Statistician	“We need to freeze μ , σ if we’re going to compare runs.”
Computer Vision Engineer	“Add a reproducibility flag or seed.”
Cognitive Scientist	“Of course the perceptual field shifts, context defines salience.”
Artist–Researcher	“That shifting is the point; it mirrors how the eye recalibrates tension.”

Provenance

Attach metadata to each run:

```
"APE_reference": {
  "mean": 3.42,
  "std": 0.56,
  "bins": [2.0, 2.8, 3.4, 3.9, 4.5],
  "mode": "dynamic"
}
```

For reproducibility, store reference μ , σ , and bin edges in a JSON log (e.g., `vcli_reference_stats.json`). That way, even if it’s dynamic, any later user can reconstruct the exact contextual field.

Summary

- A researcher expects contextual transforms to be distribution-aware.
- An engineer expects a way to freeze them.
- The correct design choice is **a dual-mode system with stored provenance**.

APE and RCE *alive*, they’re conceptually and mathematically present, but use the toggle and log their reference stats.

Appendix 2: How to stabilize runs

If you want reproducibility across experiments:

1. Freeze the reference distribution

- Compute global μ , σ , and percentile edges once from a canonical batch.
- Save as `vcli_reference_stats.json`.

2. Always transform new batches using that reference

```
z = (x - mu_ref) / sigma_ref
x_warped = np.sign(z) * (abs(z) ** gamma)
```

3. Fix bin edges for RCE

- Use constant quantile thresholds (e.g., `[0, .2, .4, .6, .8, 1.0]`) rather than dynamic `pd.qcut`.
- That way, bins represent fixed coherence bands rather than data-dependent quantiles.

4. Lock image order and count

- When comparing runs visually (APE/RCE plots), always use identical input arrays so the reference distribution and bin membership stay consistent.

Appendix 3: Empirical extension / Future forward:

Add “Empirical Validation and Drift Behavior” to show what happens numerically:

- mention that APE reduces kurtosis of mid-band distributions ($\Delta k \approx -0.4$)

- RCE increases within-band variance ($\Delta\sigma^2 \approx +0.2-0.3$)
This would be ideal for a *developer* appendix or Colab companion.

Potential future metric:

Introduce *APE-residual energy* (E_{ar}) as a scalar:

$$E_{ar} = \sum_i (y'_i - y_i)^2$$

A single-number measure of how much a dataset expands under perceptual warp. Useful for quantifying “field tension.”

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How these map:

- **Anisotropic Power Equalization (APE)**
→ Grounded by Naka–Rushton response curves; divisive normalization (Carandini & Heeger); classic γ /tone-mapping (Reinhard; Schlick); variance-stabilizing transforms (Box–Cox).
- **Residual Contrast Expansion (RCE)**
→ Within-band (SCI) local variance amplification consistent with local contrast enhancement/CLAHE (Pizer; Zuiderveld) and quantile/percentile reweighting (Bolstad et al.).
- **Batch-relative behavior**
→ Quantile normalization & order-statistics literature (Bolstad; Tukey) explain why adding/removing images shifts the non-linear mapping.
- **Perceptual alignment**
→ Psychophysical scaling (Fechner, Stevens) and color appearance models (CIE, Fairchild) justify moving from raw geometric distances to perceptual/normalized distances.

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